Name : Priyanshi Dawar
Enrollment no. : 0801IT231099
IT 2nd year A3

LAB ASSIGNMENT-01

Q1. Write a program to print largest number among the given number in an array using dynamic array.

Code :

```
#include <stdio.h>
#include <stdlib.h>

int main() {
   printf("Name : Priyanshi Dawar\n");
   printf("Enrollment No. : 0801IT231099\n");

   int n;

   // Ask the user for the number of elements
   printf("Enter the number of elements in the array: ");
   scanf("%d", &n);

   // Dynamically allocate memory for n integers
   int *arr = (int *)malloc(n * sizeof(int));

   // Check if memory allocation was successful
   if (arr == NULL) {
      printf("Memory allocation failed!\n");
      return 1;
   }

   // Input elements into the array
   printf("Enter %d elements:\n", n);
   for (int i = 0; i < n; i++) {
      printf("Element %d: ", i + 1);
```

```
        scanf("%d", &arr[i]);
    }

    // Find the largest number
    int largest = arr[0]; // Assume the first element is the largest
    for (int i = 1; i < n; i++) {
        if (arr[i] > largest) {
            largest = arr[i]; // Update largest if current element is larger
        }
    }

    // Print the largest number
    printf("The largest number in the array is: %d\n", largest);

    // Free the allocated memory
    free(arr);

    return 0;
}
```

Output:

```
/tmp/vYOd9roUSn.o
Name : Priyanshi Dawar
Enrollment No. : 0801IT231099
Enter the number of elements in the array: 4
Enter 4 elements:
Element 1: 5
Element 2: 9
Element 3: 1
Element 4: 0
The largest number in the array is: 9


=== Code Execution Successful ===
```

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Q2. Write a program to print smallest number among the given number in an array using dynamic array.

Code :

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    printf("Name : Priyanshi Dawar\n");
    printf("Enrollment No. : 0801IT231099\n");

    int n;

    // Ask the user for the number of elements
    printf("Enter the number of elements in the array: ");
    scanf("%d", &n);

    // Dynamically allocate memory for n integers
    int *arr = (int *)malloc(n * sizeof(int));

    // Check if memory allocation was successful
    if (arr == NULL) {
        printf("Memory allocation failed!\n");
        return 1;
    }

    // Input elements into the array
    printf("Enter %d elements:\n", n);
    for (int i = 0; i < n; i++) {
        printf("Element %d: ", i + 1);
        scanf("%d", &arr[i]);
    }
```

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```
    // Find the smallest number
    int smallest = arr[0]; // Assume the first element is the smallest
    for (int i = 1; i < n; i++) {
        if (arr[i] < smallest) {
            smallest = arr[i]; // Update smallest if current element is smaller
        }
    }

    // Print the smallest number
    printf("The smallest number in the array is: %d\n", smallest);

    // Free the allocated memory
    free(arr);

    return 0;
}
```

Output:

```
/tmp/XBdmhMFEvc.o
Name : Priyanshi Dawar
Enrollment No. : 0801IT231099
Enter the number of elements in the array: 4
Enter 4 elements:
Element 1: 2
Element 2: 0
Element 3: -4
Element 4: 6
The smallest number in the array is: -4


=== Code Execution Successful ===
```

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Q3. Write a program to print the elements of array in reverse order. (show the use of malloc(), realloc() and free() function in your program.)

Code:

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    printf("Name : Priyanshi Dawar\n");
    printf("Enrollment No. : 0801IT231099\n");

    int n, new_size;

    // Ask the user for the number of elements in the array
    printf("Enter the number of elements in the array: ");
    scanf("%d", &n);

    // Dynamically allocate memory using malloc()
    int *arr = (int *)malloc(n * sizeof(int));

    // Check if malloc() was successful
    if (arr == NULL) {
        printf("Memory allocation failed!\n");
        return 1;
    }

    // Input elements into the array
    printf("Enter %d elements:\n", n);
    for (int i = 0; i < n; i++) {
        printf("Element %d: ", i + 1);
        scanf("%d", &arr[i]);
    }

    // Reallocate memory to change the array size (optional step to show use of realloc)
```

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```
    printf("\nWould you like to resize the array? Enter the new size (Enter the same size if no resize): ");
    scanf("%d", &new_size);

    // Resize the array using realloc()
    arr = (int *)realloc(arr, new_size * sizeof(int));

    // Check if realloc() was successful
    if (arr == NULL) {
        printf("Memory reallocation failed!\n");
        return 1;
    }

    // If the new size is larger, ask the user to input additional elements
    if (new_size > n) {
        printf("Enter %d more elements:\n", new_size - n);
        for (int i = n; i < new_size; i++) {
            printf("Element %d: ", i + 1);
            scanf("%d", &arr[i]);
        }
    }

    // Print elements in reverse order
    printf("\nArray elements in reverse order:\n");
    for (int i = new_size - 1; i >= 0; i--) {
        printf("%d ", arr[i]);
    }
    printf("\n");

    // Free the allocated memory
    free(arr);

    return 0;
}
```

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Output:

```
/tmp/VnwcF8mV8t.o
Name : Priyanshi Dawar
Enrollment No. : 0801IT231099
Enter the number of elements in the array: 4
Enter 4 elements:
Element 1: 4
Element 2: 8
Element 3: 1
Element 4: 0

Would you like to resize the array? Enter the new size (Enter the same size
    if no resize): 6
Enter 2 more elements:
Element 5: 2
Element 6: 7

Array elements in reverse order:
7 2 0 1 8 4


=== Code Execution Successful ===
```

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Q4. Write a program to find the occurrence of duplicate element in the array.

Code:

```
#include <stdio.h>

int main() {
    printf("Name : Priyanshi Dawar\n");
    printf("Enrollment No. : 0801IT231099\n");

    int n;

    // Ask the user for the number of elements in the array
    printf("Enter the number of elements in the array: ");
    scanf("%d", &n);

    int arr[n];
    int count[n];

    // Input elements into the array
    printf("Enter %d elements:\n", n);
    for (int i = 0; i < n; i++) {
        printf("Element %d: ", i + 1);
        scanf("%d", &arr[i]);
        count[i] = -1; // Initialize count array with -1
    }

    // Find duplicate elements
    for (int i = 0; i < n; i++) {
        int occurrence = 1;
        for (int j = i + 1; j < n; j++) {
            if (arr[i] == arr[j]) {
                occurrence++;
                count[j] = 0; // Mark duplicates to avoid rechecking
```

```
}
        }
        if (count[i] != 0) {
            count[i] = occurrence;
        }
    }

    // Print the duplicate elements and their occurrences
    printf("\nDuplicate elements and their occurrences:\n");
    int foundDuplicates = 0;
    for (int i = 0; i < n; i++) {
        if (count[i] > 1) {
            printf("%d occurs %d times\n", arr[i], count[i]);
            foundDuplicates = 1;
        }
    }

    if (!foundDuplicates) {
        printf("No duplicate elements found.\n");
    }

    return 0;
}
```

Output:

```
/tmp/mEMzDh835H.o
Name : Priyanshi Dawar
Enrollment No. : 0801IT231099
Enter the number of elements in the array: 4
Enter 4 elements:
Element 1: 5
Element 2: 3
Element 3: 4
Element 4: 5

Duplicate elements and their occurrences:
5 occurs 2 times


=== Code Execution Successful ===
```

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Q5. Write a program to take array as input from user and find out sum of odd and even.

Code :

```
#include <stdio.h>
#include <stdlib.h> // Required for malloc and free

int main() {
    int n;

    printf("Name : Priyanshi Dawar\n");
    printf("Enrollment No. : 0801IT231099\n");


    // Input the number of elements in the array
    printf("Enter the number of elements: ");
    if (scanf("%d", &n) != 1) {
        printf("Invalid input\n");
        return 1;
    }

    // Allocate memory for the array
    int *array = (int *)malloc(n * sizeof(int));

    // Check if malloc was successful
    if (array == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    // Input the elements of the array
    printf("Enter %d integers:\n", n);
    for (int i = 0; i < n; i++) {
        if (scanf("%d", &array[i]) != 1) {
            printf("Invalid input\n");
```

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```
        free(array); // Free allocated memory before exit
            return 1;
        }
    }

    // Initialize sums
    int sumEven = 0;
    int sumOdd = 0;

    // Calculate sums of even and odd numbers
    for (int i = 0; i < n; i++) {
        if (array[i] % 2 == 0) {
            sumEven += array[i];
        } else {
            sumOdd += array[i];
        }
    }

    // Print the results
    printf("Sum of even numbers: %d\n", sumEven);
    printf("Sum of odd numbers: %d\n", sumOdd);

    // Free the allocated memory
    free(array);

    return 0;
}
```

Output :

```
PS C:\Users\dawar\OneDrive\Tài liệu\college study material\3rd sem\Assignments\DS\assignment 01> gcgcc user.c
PS C:\Users\dawar\OneDrive\Tài liệu\college study material\3rd sem\Assignments\DS\assignment 01> gc./a.exe
Name : Priyanshi Dawar
Enrollment No. : 0801IT231099
Enter the number of elements: 4
Enter 4 integers:
1 6 1 9
Sum of even numbers: 6
Sum of odd numbers: 11
```

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Q6. Write a program to print data of an array in ascending order.

Code:

```
#include <stdio.h>

int main() {
    printf("Name : Priyanshi Dawar\n");
    printf("Enrollment No. : 0801IT231099\n");

    int n, temp;

    // Ask the user for the number of elements in the array
    printf("Enter the number of elements in the array: ");
    scanf("%d", &n);

    int arr[n];

    // Input elements into the array
    printf("Enter %d elements:\n", n);
    for (int i = 0; i < n; i++) {
        printf("Element %d: ", i + 1);
        scanf("%d", &arr[i]);
    }

    // Bubble Sort algorithm to sort the array in ascending order
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                // Swap arr[j] and arr[j + 1]
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
```

```
        }
    }

    // Print the sorted array in ascending order
    printf("\nArray in ascending order:\n");
    for (int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");

    return 0;
}
```

Output:

```
/tmp/U6V1ekKM8z.o
Name : Priyanshi Dawar
Enrollment No. : 0801IT231099
Enter the number of elements in the array: 5
Enter 5 elements:
Element 1: 4
Element 2: 8
Element 3: 1
Element 4: 3
Element 5: 6

Array in ascending order:
1 3 4 6 8


=== Code Execution Successful ===
```